



USED CAR PRICE PREDICTION

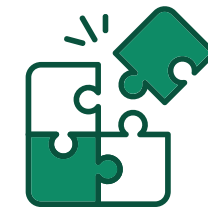
Presented by: Rimberio Group

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Problem Definition

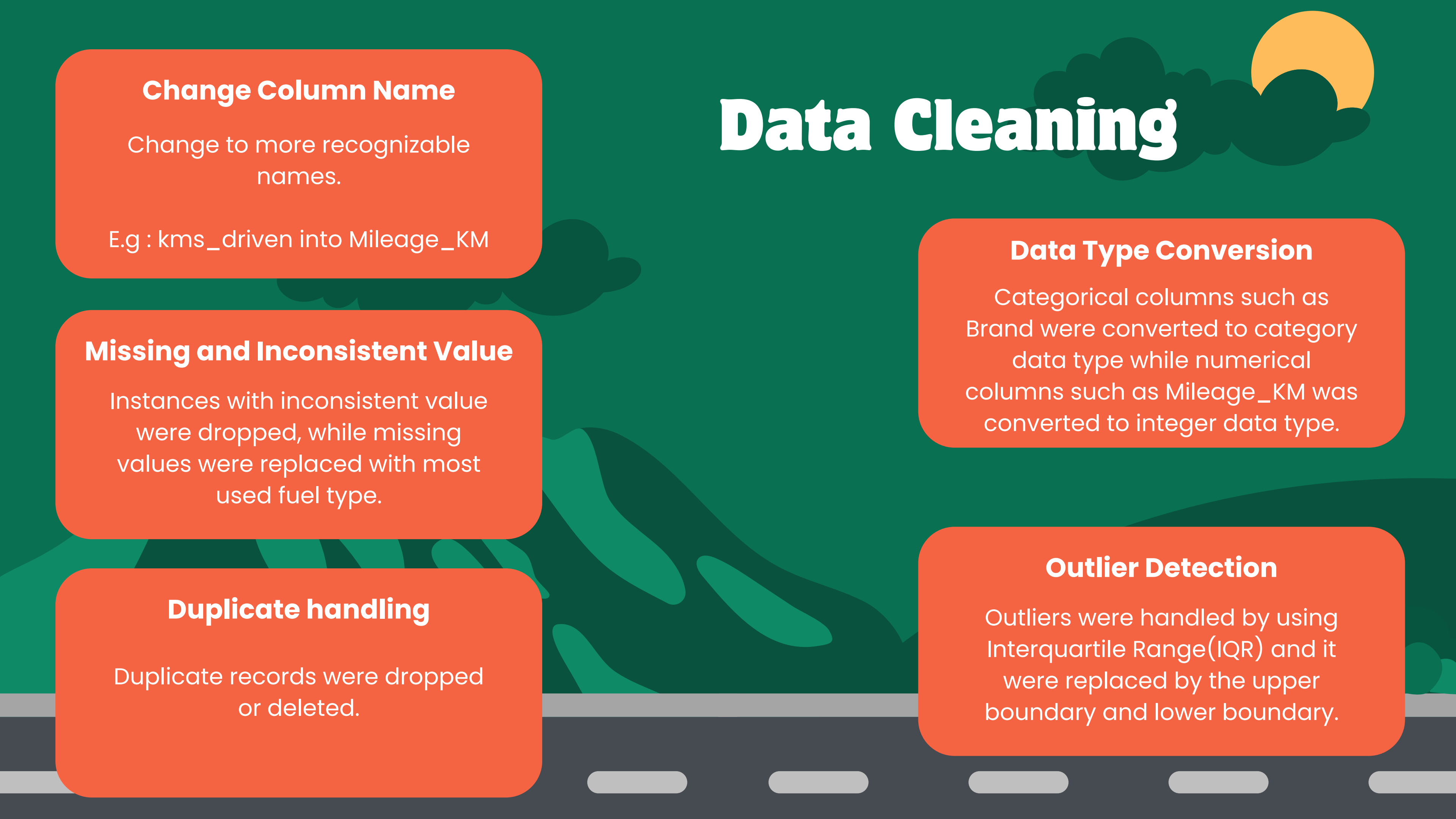
In this generation where there are billions of people in this world, some people prefer to buy a used car. Therefore the used car market has been highly in demand that been affected by the physical and economic factors. Unlike new car, the value of used car has been varies over the time.

Platform like Quikr has been manage thousands of active listings of the used car, creating a vast digital market. However, because of the market is unstructured, a significant challenge both consumer and automotive platforms.



DATASET DESCRIPTION

Variables	Type	Variable	Description
name	Categorical	Independent Variable	The model name of a car
company	Categorical	Independent variable	The manufacturing company of the car
year	Numerical	Independent Variable	The model year or year of manufacture of the vehicle
Price	Numerical	Dependent Variable	The resale price of the car (RM)
kms_driven	Numerical	Independent Variable	The total distance accumulated by the car in kilometers
fuel_type	Categorical	Independent Variable	The engine fuel classification (e.g Petrol, Diesel)



Data Cleaning

Change Column Name

Change to more recognizable names.

E.g : kms_driven into Mileage_KM

Missing and Inconsistent Value

Instances with inconsistent value were dropped, while missing values were replaced with most used fuel type.

Duplicate handling

Duplicate records were dropped or deleted.

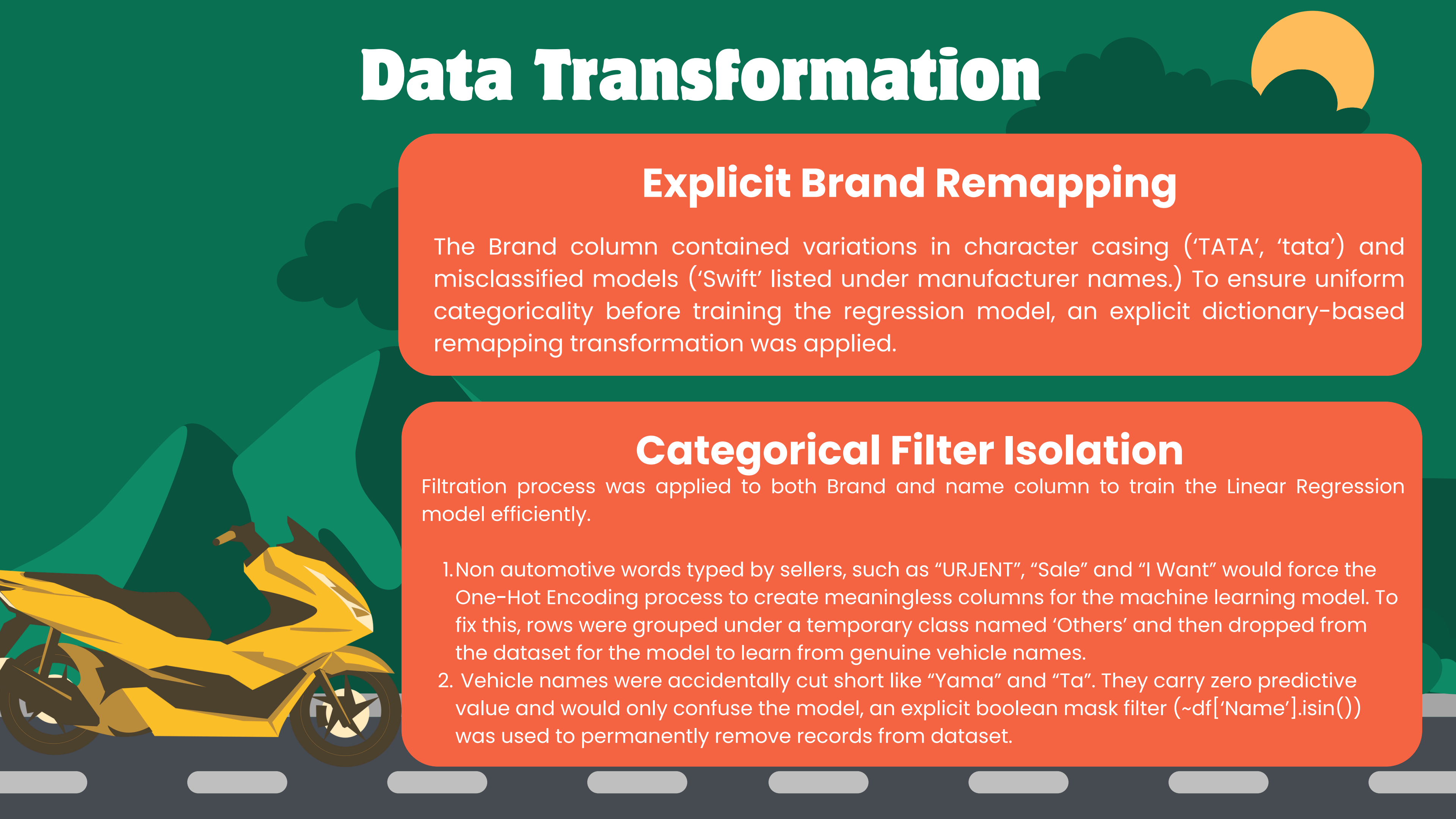
Data Type Conversion

Categorical columns such as Brand were converted to category data type while numerical columns such as Mileage_KM was converted to integer data type.

Outlier Detection

Outliers were handled by using Interquartile Range(IQR) and it were replaced by the upper boundary and lower boundary.

Data Transformation



Explicit Brand Remapping

The Brand column contained variations in character casing ('TATA', 'tata') and misclassified models ('Swift' listed under manufacturer names.) To ensure uniform categoricity before training the regression model, an explicit dictionary-based remapping transformation was applied.

Categorical Filter Isolation

Filtration process was applied to both Brand and name column to train the Linear Regression model efficiently.

1. Non automotive words typed by sellers, such as "URJENT", "Sale" and "I Want" would force the One-Hot Encoding process to create meaningless columns for the machine learning model. To fix this, rows were grouped under a temporary class named 'Others' and then dropped from the dataset for the model to learn from genuine vehicle names.
2. Vehicle names were accidentally cut short like "Yama" and "Ta". They carry zero predictive value and would only confuse the model, an explicit boolean mask filter (`~df['Name'].isin()`) was used to permanently remove records from dataset.

RESULTS & FINDINGS



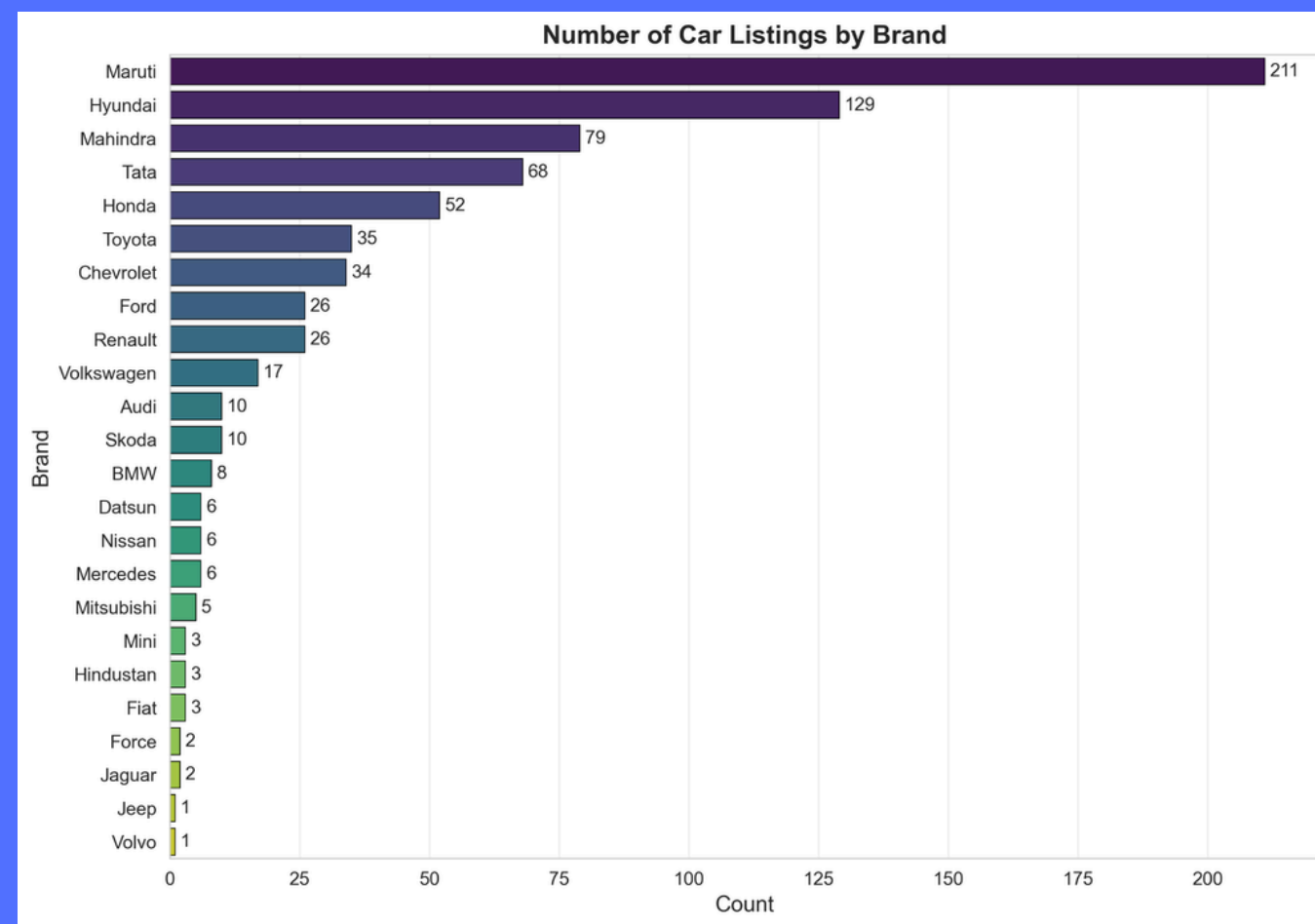
1 Descriptive Statistical Analysis

Table 4.1.1 : Descriptive Statistics Matrix for Used Car Price Prediction Dataset

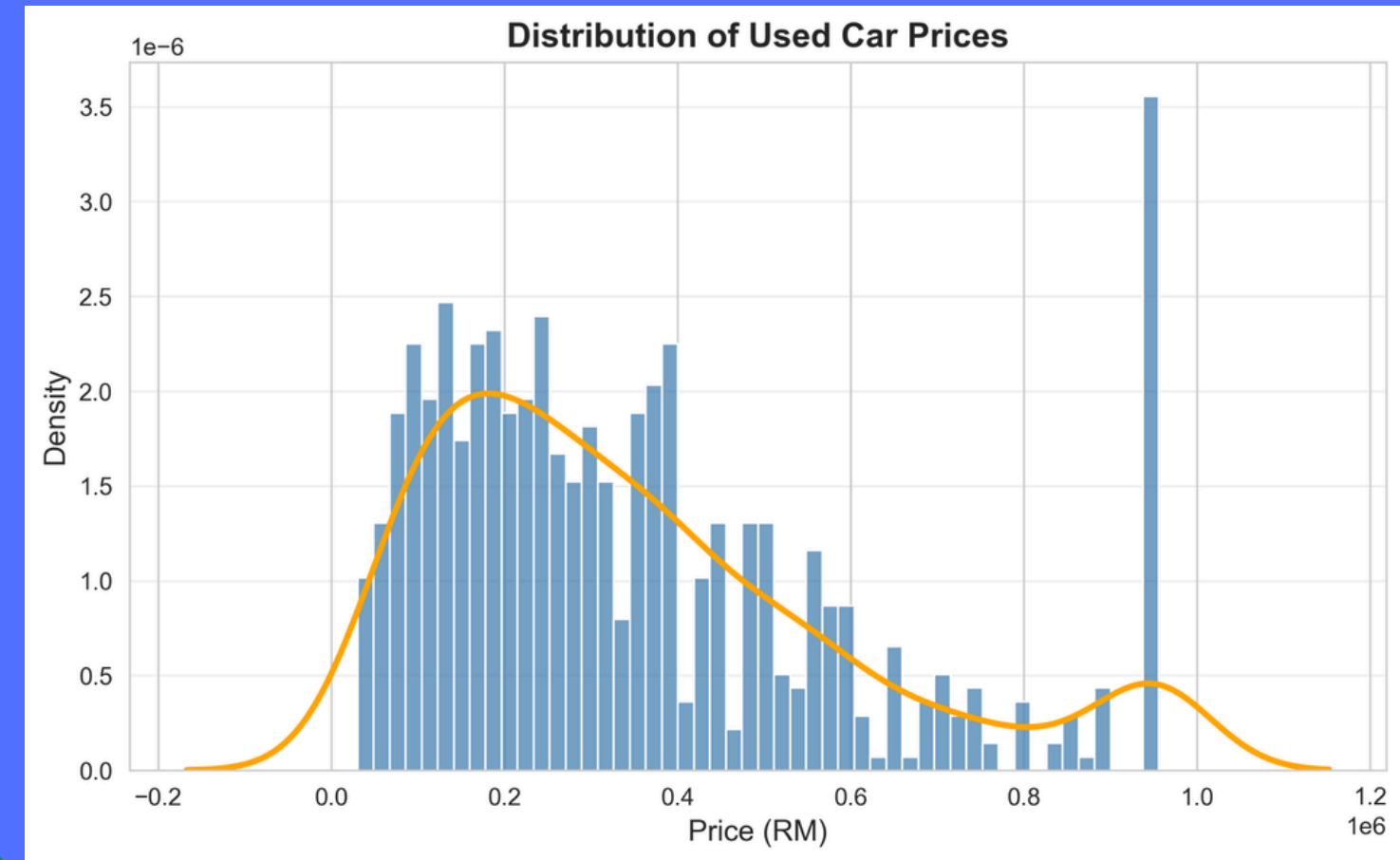
Column	Mean	Median	Standard Deviation	Minimum	Maximum
Price_RM	354 911.46	290 000.00	246 524.24	30 000.00	95 5625.00
Mileage_KM	44 273.14	41 000.00	25 790.84	0.00	103 000.00
Year	2012	2013	4.07	1995	2019

2 Data Visualisation

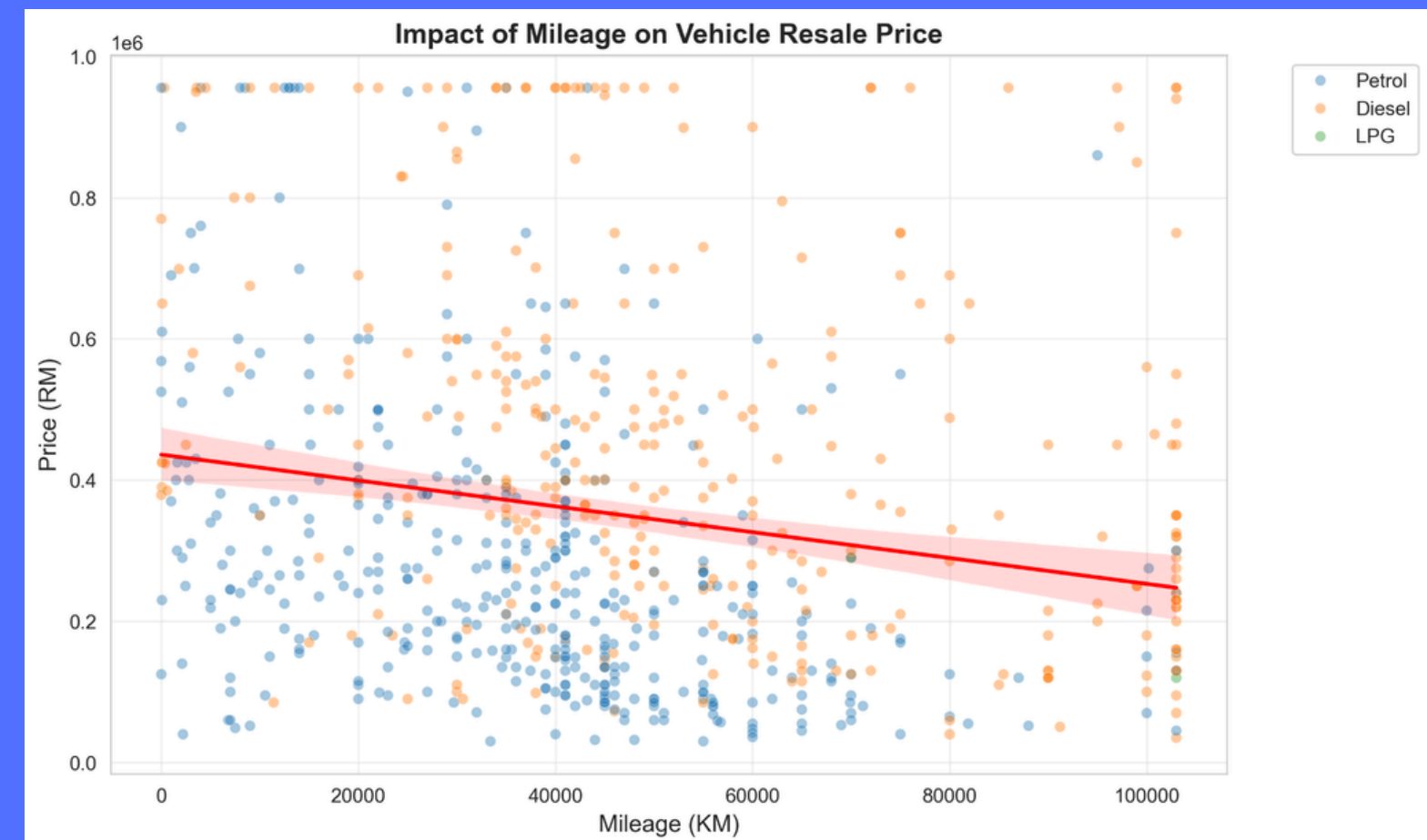
Brand Distribution



Price Distribution



Mileage-Price Distribution



Regression Model

- To build a model to analyse the relationship between used car characteristics and the resale price
- Target Variable: Price_RM

Model Choice

- Regression model selected to predict continuous outcome
- Linear regression chosen as the baseline programming

Why Linear Regression?

- Provides a simple, interpretable starting point.
- Helps understand how characteristics influence resale price.

Model Evaluation

- Using 70% for training slice
- Using 30% for testing slice

• R-squared (R² Score)

Measure the proportion of the variance in the car price that can be successfully predicted by the features provided

• Mean Absolute Error (MAE)

Calculate the absolute difference between the model's predicted car prices and the actual prices

• Root Mean Squared Error (RMSE)

Calculates the square root of the average square error



Results

Evaluation Metric (Linear Regression)	Experimental Result
Training Sample 70%	520 rows
Testing Sample 30%	223 rows
Mean Absolute Error (MAE)	RM 123976.29
Root Mean Squared Error (RMSE)	RM 171161.99
R-squared (R^2 Score)	0.5382

Teamwork Evaluation

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Communication

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Team Performance

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Conclusion & Recommendation

Key Findings

- ✓ Affordable brands dominate the used car market
- ✓ Mileage and fuel type significantly influence vehicle prices
- ✓ Linear Regression achieved moderate performance ($R^2 = 0.5382$)

Insights

- Automated price estimates have limitations
- Data quality issues affect prediction reliability
- Price explainability improves user trust

Recommendations

- ✓ Include more vehicle-related features
- ✓ Improve outlier handling and data validation
- ✓ Explore advanced models
 - Random Forest
 - XGBoost



Thank you

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